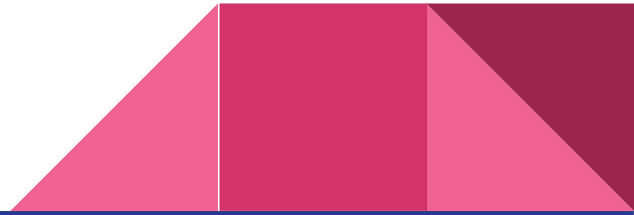
Project INSPIRE Meeting II

Liam Gallagher

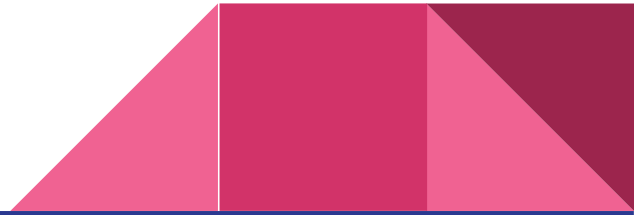
Agenda

- System Design Principles
- LC2/Signal Background
- Simulation
- Group Update



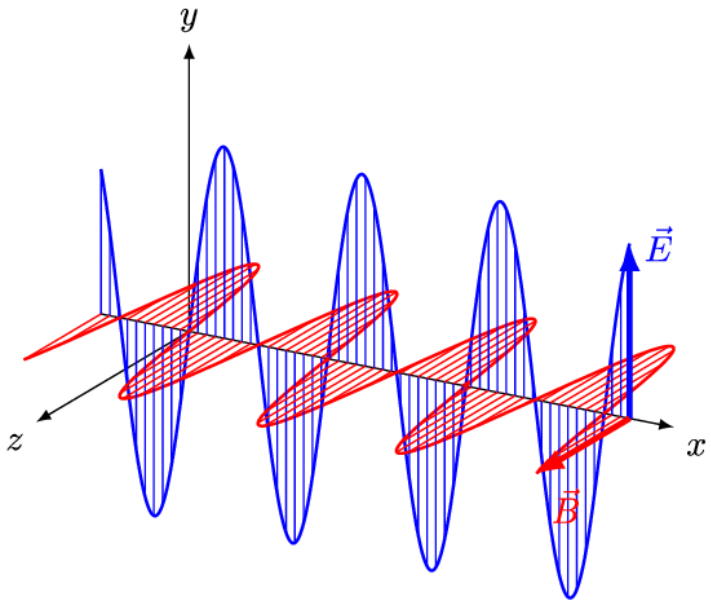
System Design Principles

- All projects start with a goal!
- We want to digitize and record electromagnetic waves in the 3-30kHz range
- Let’s work our way through the system.



Electromagnetic Waves (a Quick Refresher)

- Caused by movement of charged particles that induces a wave in both E-field and H-field
- Can transmit energy and information
- Will propagate/move through all materials
  - Differing levels of effectiveness
- Picked up by antennas



E-field vs H-field Antennas

- Two major families of near field antennas: E-field probes and H-field probes
  - Each focuses on one component of wave
- E-field probes measure alternating voltage potentials
- H-field probes measure magnetic flux

E-field Antennas

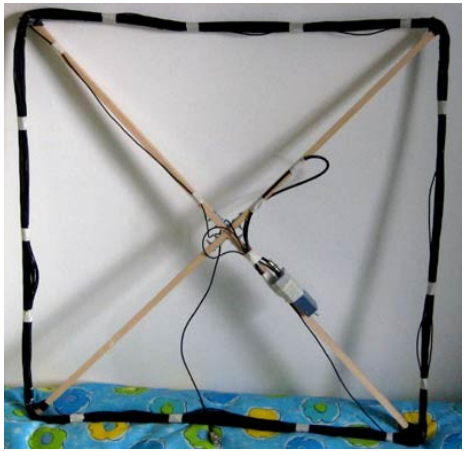
Advantages	Disadvantages
• Very sensitive • Smaller Size • Lower Cost	• No directionality • Vulnerable to interference





H-field Antennas

Advantages	Disadvantages
• Directional! • If 2x or more are made • Resistant to constant interference	• Big size • Higher cost





Antenna Design

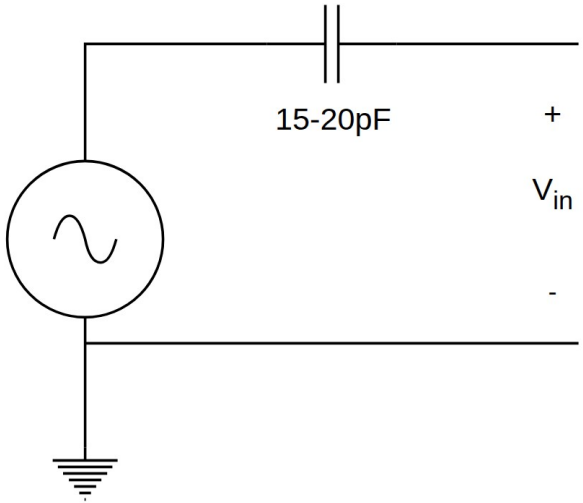
- As such, we are using a E-field probe. This saves us cost in both material (not having to tear up a bunch of cable to make a mile-long spool) and in complexity (single antenna front-end).

- A extending whip antenna is being chosen.



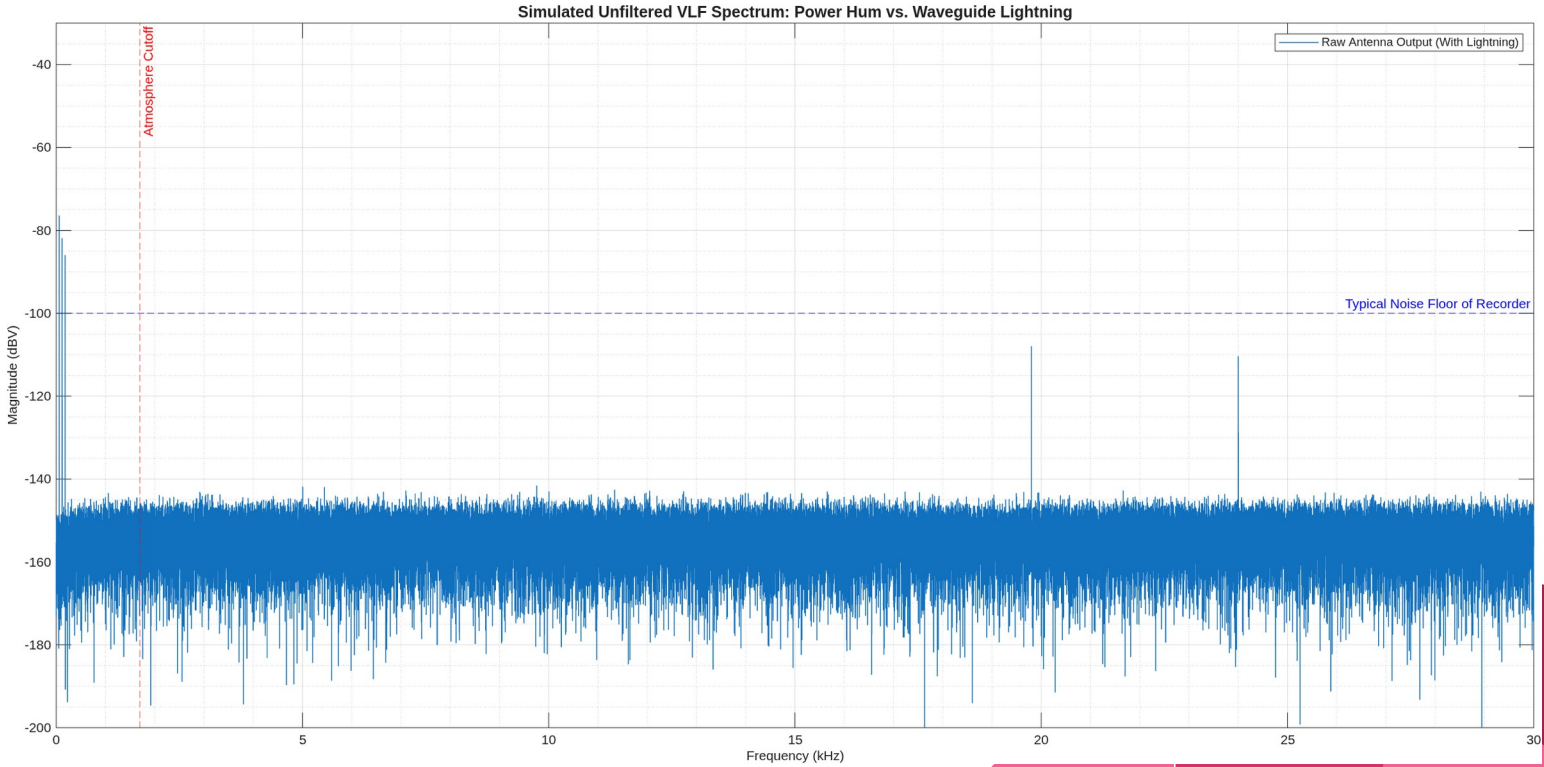
Issues? Already?

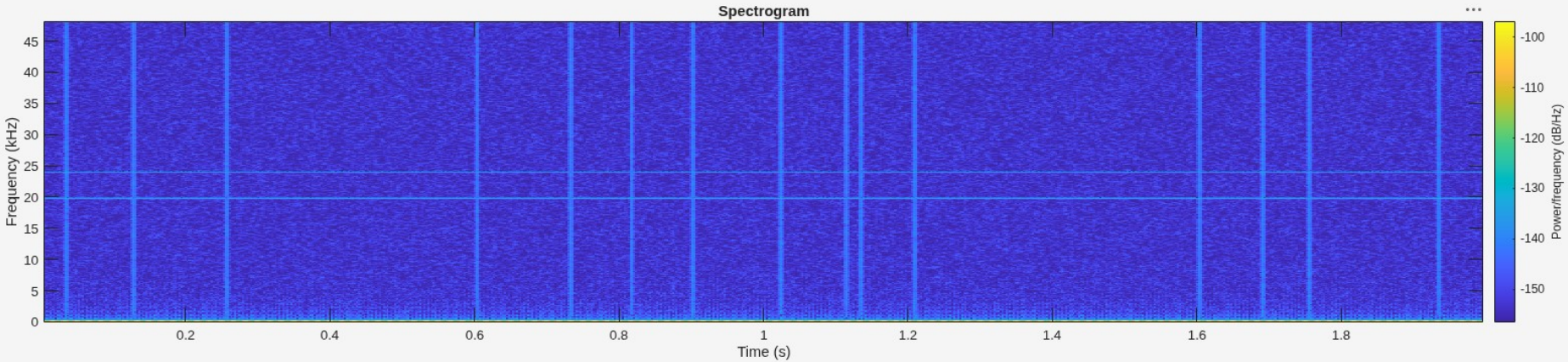
- No free lunches in engineering, there are issues already.
- These E-field whips behave as extremely small AC sources with a built in tiny capacitance.
- Hooking these directly into something to measure requires nanovolt accuracy.
- This signal is also broadband!

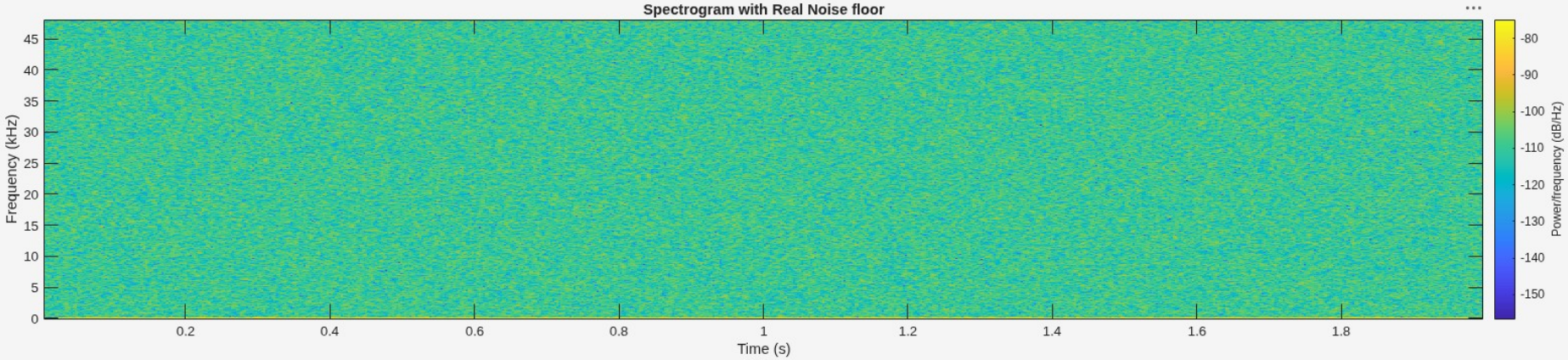


Simulated Direct Spectrum

https://www.falstad.com/fourier/

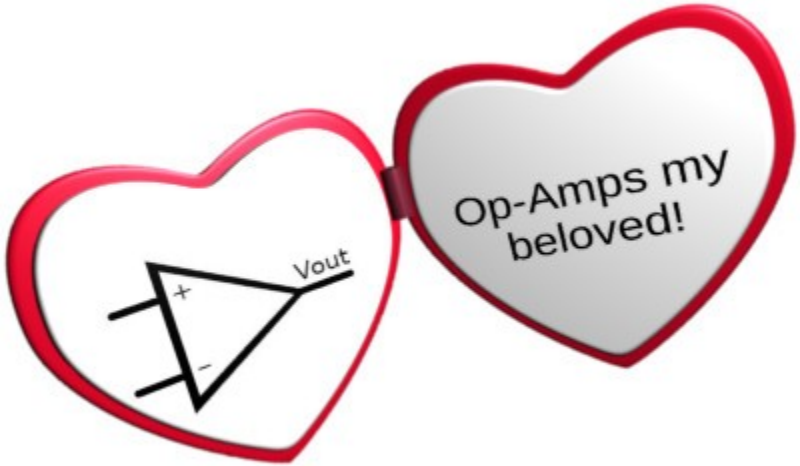



Spectrogram
Frequency (kHz)
0
5
10
15
20
25
30
35
40
45
0.2
0.4
0.6
0.8
1
1.2
1.4
1.6
1.8
Time (s)
-100
-110
-120
-130
-140
-150
Power/frequency (dB/Hz)


Spectrogram with Real Noise floor
Frequency (kHz)
0
5
10
15
20
25
30
35
40
45
0.2
0.4
0.6
0.8
1
1.2
1.4
1.6
1.8
Time (s)
-80
-90
-100
-110
-120
-130
-140
-150
Power/frequency (dB/Hz)

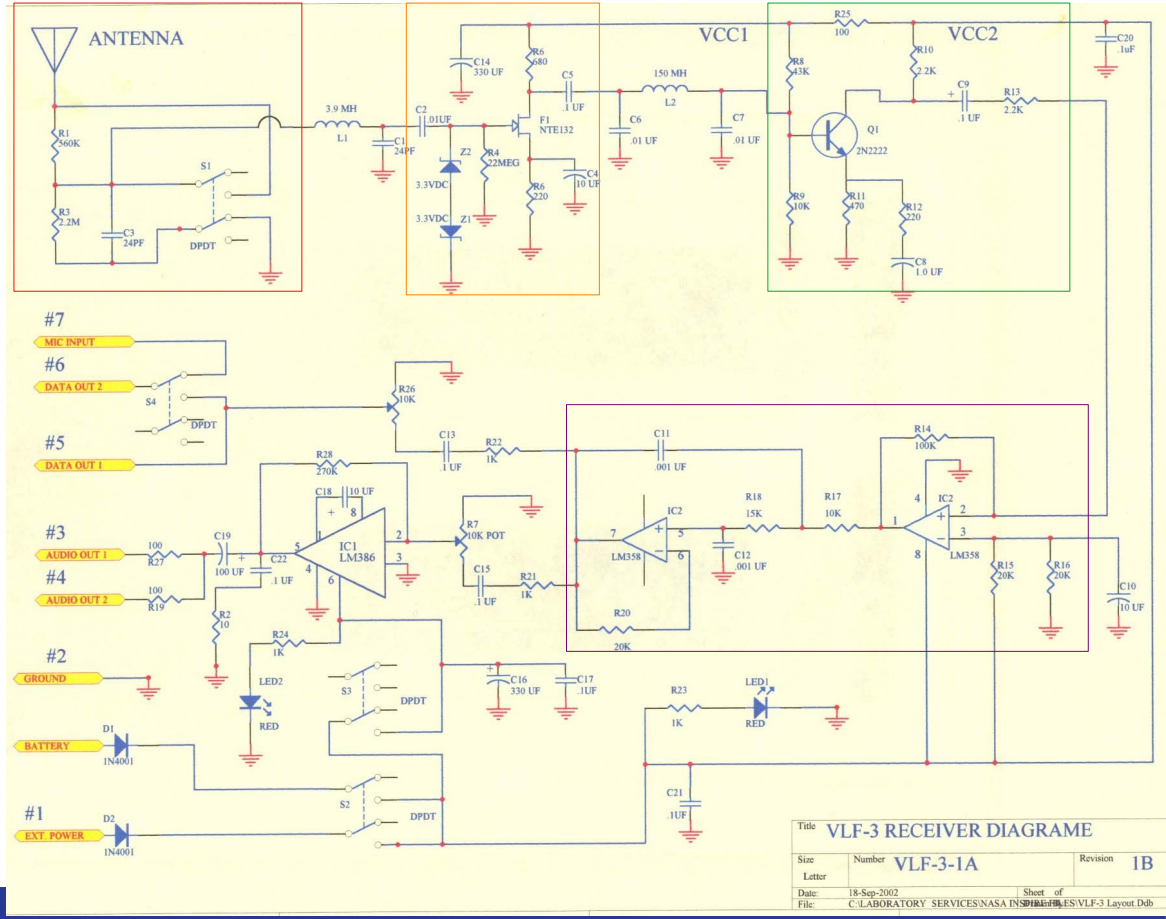
Our Solution

- Amplification and Filtering are our solutions!
- Practically, each section of the analog front-end is either amplifying the signal (increasing amplitude), filtering the signal (removing frequency components), or both.
- Operational Amplifiers make this work far easier and more consistent.

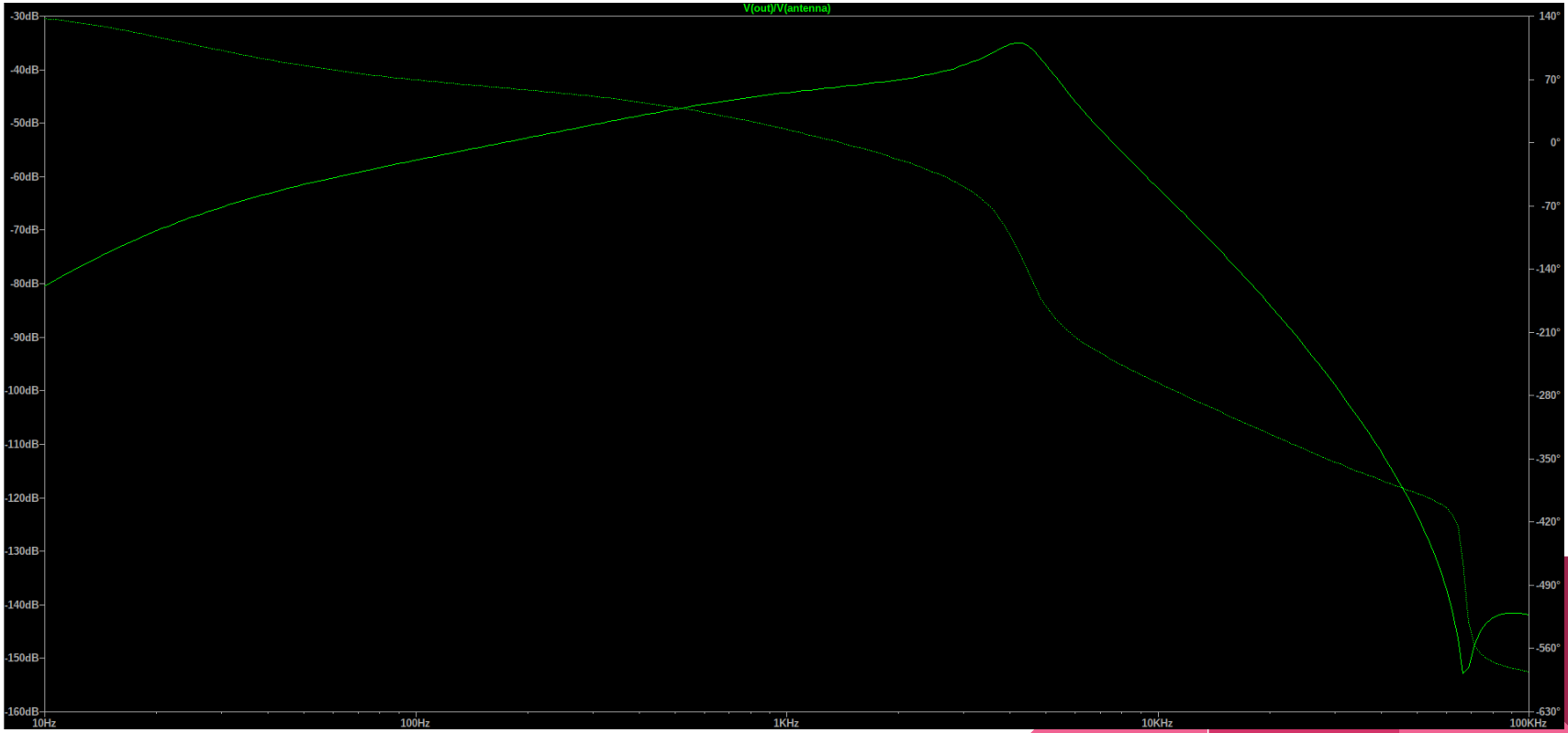


VLF Schematic

- Four main sections
- Red – Antenna
- Orange – Impedance Matching Circuit
  - Done by common-drain amplifier
- Green – Amplifier
- Black – Amplifier and Active Filter
- Inductor-Capacitor LC filters between stages
- LM386 – Audio headphone amplifier



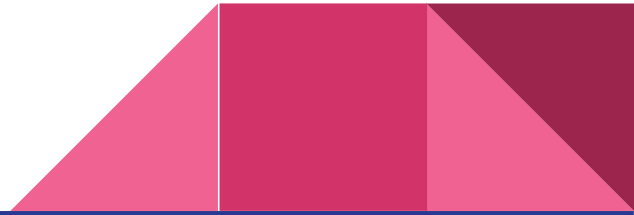
LTSpice Results



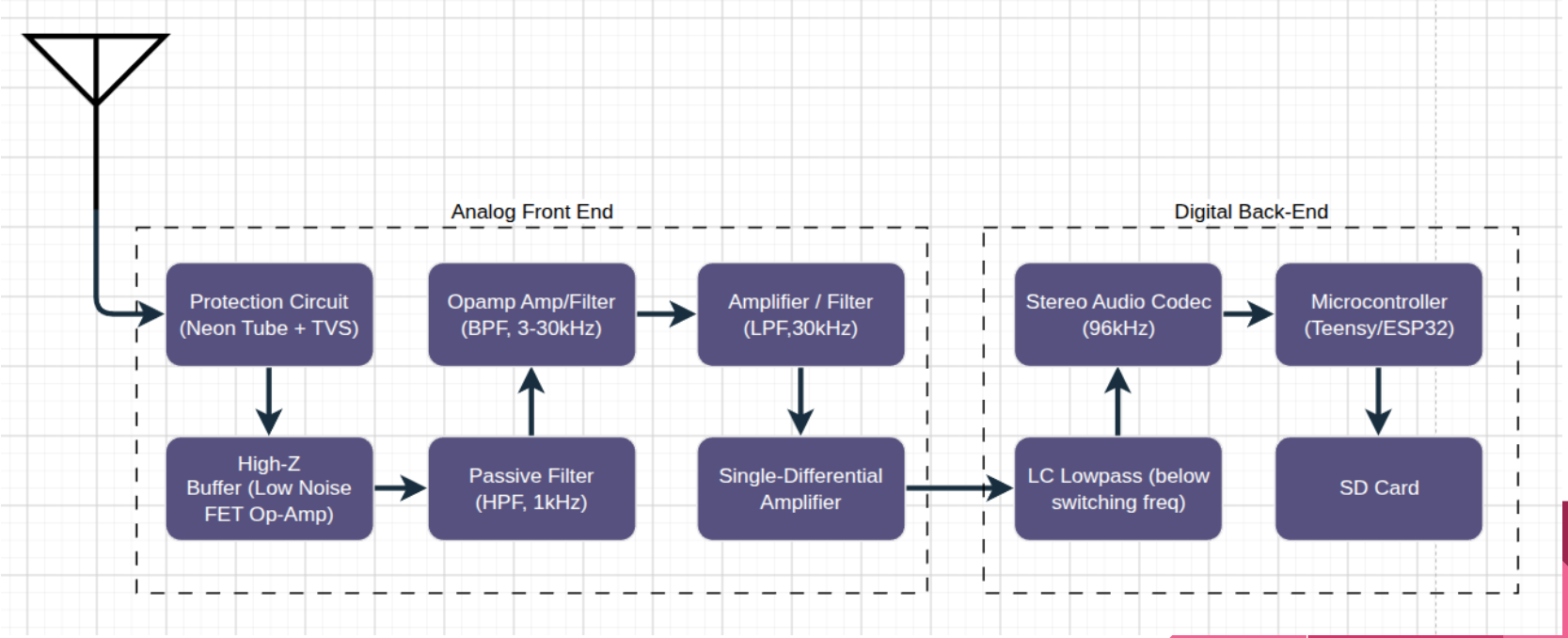
Analysis

- The trade-off made by the NASA engineers was to sacrifice gain and bandwidth in order to ensure rejection of the power line harmonics and other LORAN transmitters.

- Our design will take advantage of modern low-noise high-bandwidth op-amps.



Design Block Diagram



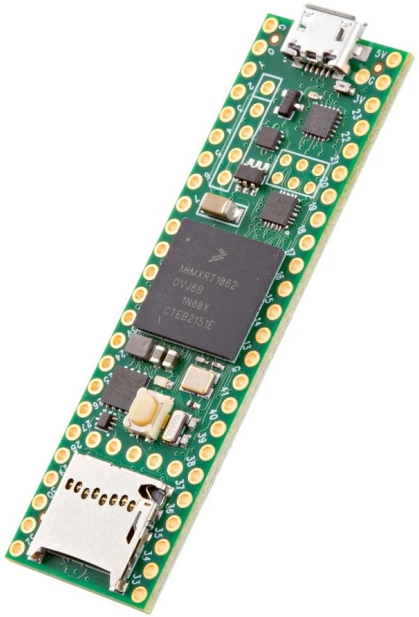
Differential

- Why differential?
  - Differential signal routing reduces the effects of noise, because noise sources picked up by both runs of the cable will be removed
- XLR is an audio cable standard that routes differential audio (also known as “balanced”) over long distances, which is perfect for our needs!



MCU Choice

- MCU Choice is an incredibly controversial topic, with different EEs preferring various manufacturers.
- For this use case, a flexible, easy to work with MCU is essential.
- Teensy 4.1 provides the fastest MCU for its price bracket, Arduino IDE support + existing audio toolchain, SD card slot built in



Group Updates

- Currently, out of the 32 respondents to the survey, 5 preferred the digital side, 14 preferred the analog side, and 13 said they would be fine either way.

- This is totally okay! I will lean the meeting teaching topics more towards the analog circuitry side, and I can work with the digital team to assist in the digital PCB design.

Very Rough Timeline

- By next meeting, I will have analog and digital teams set, channels on IEEE discord should be made.
- Analog and Digital subleads will be selected and I will talk with them about how they want to assign people to each sub-circuit.
- Hopefully by mid- to late- October, we can have circuit simulations done or near completion.

- Bread-boarding/Prototyping can happen following completed simulations.

- Two semester project, we don’t have to rush, but we do need to keep progress steady!

What do I do now?

- I will be posting a technical writeup by the NASA engineers on how the VLF-3 INSPIRE receiver works, take a look at it!
- If you know LTSpice, start simulating amplifiers or filters! We can start with idealized components, and incorporate real component models as we discuss them.
- If you don’t know LTSpice, that is okay! Analog Devices (the developer) have a 4-video series, and this is good to watch to learn the basics. (It will be **VERY** helpful for labs in your later classes)

